



KALACHAKRA

A forecasting model, and the instrument that judged it

Companion to EKAGRA (SIH26145)

The question nobody can answer today

A classifier scores the traffic it observed. It cannot represent an action that was never taken, so it cannot answer the question a defender actually asks: **what happens if I isolate this host now?** KALACHAKRA learns the transition dynamics of a host-interaction graph and rolls them forward six steps with no new observations, which makes that question expressible.

Our headline claim is refuted, by our own experiments

We predicted a world model would warn earlier than a classifier. It does not. On **CTU-13** - thirteen real botnet captures, 19.9 million flows, leave-one-scenario-out - it catches 5 of 12 infections at a 10% false-alarm budget. A plain gradient-boosted tree catches 7 to 9, and logistic regression still wins AUC and F1.

CTU-13: thirteen real botnet captures, leave-one-scenario-out, false alarms capped at 10%

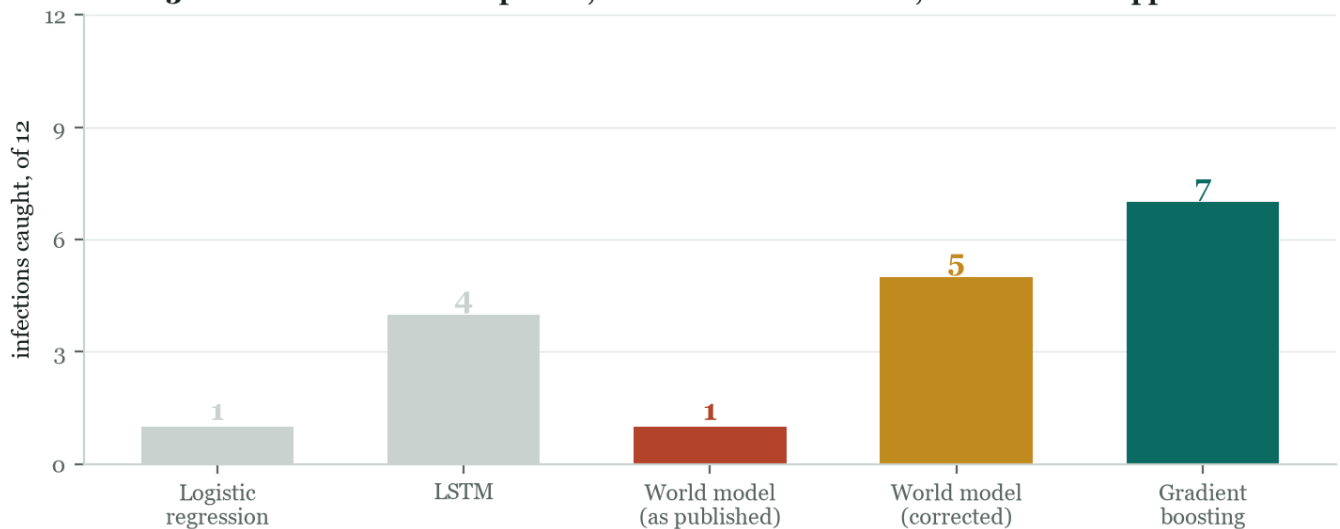


Figure 1. CTU-13, 13 scenarios, 118 episodes, 12 scored infections, false alarms capped at 10%. Sources: exp14_verdicts.json and exp16_verdicts.json.

5 of 12

infections
the world model catches

7 to 9

a gradient-boosted tree catches

16 of 40

pre-registered predictions refuted

107

automated
tests passing

We published 1 of 12. It was wrong, and finding out is the contribution

Our own published figure made us look worse than we were, and we corrected it anyway. Experiment 16 could not reproduce it; a deterministic control (gradient boosting, which reproduced the old row to the digit) proved the pipeline was identical; six seeds ruled out noise; and experiment 18 found the cause.

Training longer raises AUC and lowers detections

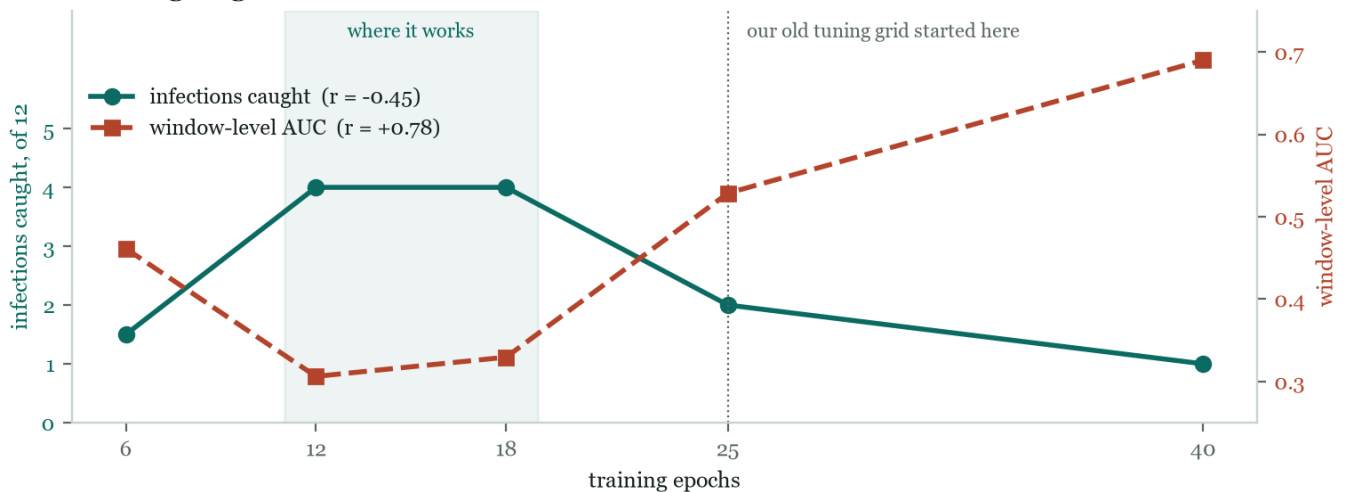


Figure 2. Training longer raises window-level AUC and lowers the thing we actually report. Source: exp18_verdicts.json.

Detections and AUC move in **opposite** directions: epochs against detections $r = -0.451$, epochs against AUC $r = +0.780$. Our earlier tuning pass selected on AUC, so it was steering away from the metric we report, and its grid began at 25 epochs - outside the region that works (12 to 18). A selection criterion that is not the reported metric needs an argument that the two agree. Ours did not.

What survives, and is worth your attention

What	Evidence
A question the field cannot pose	roll out under action=none and action=isolate from one bit-identical prefix; the gap is the model's belief about the intervention
An evaluation instrument	lead time at a <i>matched</i> false-alarm rate, leave-one-scenario-out over 13 real captures, contained attacks counted as negatives
A finding about the problem	all four methods span 0.03 to 0.99 AUC across captures: cross-network transfer is near a coin flip, for everyone
Attribution with a checkable axiom	integrated gradients, whose completeness is verified in the library, in the payload's tests, and again in the browser

What we will not claim

Claim	Status
Forecasts earlier than a classifier	Refuted on generated data and again on CTU-13
Tells you which intervention works	Refuted four ways: $r = -0.105, -0.103, +0.037, +0.011$
Direction accuracy of 94%	Not accuracy. A constant "always isolate" predictor scores 100%; the number is evidence about the LSTM's failure (48%), not about ranking
The counterfactual is validated on real data	It cannot be. No capture holds the branch that did not happen
Ready to deploy	No inference service, no streaming graph builder

Why this is worth a judge's time

Every experiment's predictions were written into its file before it ran, and the run prints the verdict; 16 of 40 failed and all 16 are published. The instrument that refuted our model is reusable by anyone working this corpus, and it caught an error in our own published number. A result that only ever confirms its authors is the one to distrust.

Going deeper: the 42-page report is [REPORT_KALACHAKRA_SIH26153.pdf](#); the console, code and every verdict are at github.com/AKRai-2005/kalachakra. Companion project EKAGRA (SIH26145) supplies 60 to 70% of this project's data foundation.